

KAKINADA, IndiaWMO: **431890**Lat: **16.950N**Lon: **82.233E**Elev: **26**StdP: **14.68**Time Zone: **5.50 (IND)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	66.4	67.8	57.1	69.6	76.5	59.4	75.8	75.3	14.2	79.3	10.4	78.8	0.4	320	0.320

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	15.6	103.2	81.4	99.8	81.7	97.4	81.5	85.1	94.5	84.3	93.5	83.5	92.3	4.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.8	171.3	90.0	82.1	167.3	89.3	81.4	163.6	88.7	49.6	95.5	48.6	93.5	47.8	92.5	93.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	11.8	9.0	7.3	62.4	110.5	1.9	3.9	61.1	113.3	60.0	115.6	58.9	117.8	57.5	120.7	
(5)				58.7	87.0	2.1	2.1	57.2	88.5	56.0	89.8	54.8	91.0	53.3	92.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.5	76.6	79.7	84.0	87.9	90.8	88.4	85.1	84.6	84.6	83.4	79.9	76.9
	DBStd	5.18	2.12	2.24	2.31	2.45	3.77	4.69	3.55	2.87	2.67	2.49	2.23	2.17
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12230	826	831	1055	1137	1264	1153	1087	1073	1037	1035	897	835
	CDD65	6755	361	411	590	687	799	703	622	608	587	570	447	370
	CDH74	79978	2538	3722	6857	9196	11603	9802	7876	7520	7139	6639	4350	2736
	CDH80	37035	565	1294	3065	4979	7177	5589	3619	3255	3007	2597	1292	596
Wind	WSAvg	2.3	1.8	1.8	1.8	2.2	2.4	3.0	3.3	3.0	2.1	1.8	2.2	1.9
Precipitation	PrecAvg	41.3	0.2	0.2	0.4	0.8	1.2	4.5	6.8	6.1	6.8	8.5	4.9	0.7
	PrecMax	58.7	1.7	1.2	6.5	6.1	8.8	8.9	11.8	14.8	18.9	22.2	25.7	5.6
	PrecMin	24.8	0.0	0.0	0.0	0.0	0.0	0.4	3.1	0.7	1.2	1.6	0.0	0.0
	PrecStd	9.3	0.4	0.4	1.3	1.3	2.1	2.3	2.5	3.7	3.9	5.9	6.2	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.9	93.4	96.3	100.1	110.1	108.5	99.9	96.1	95.9	94.9	90.4	87.7
		MCWB	74.9	73.7	78.4	81.8	78.4	80.0	79.7	80.9	80.9	79.9	76.3	74.5
	2%	DB	86.0	90.5	94.3	98.0	104.6	103.1	96.2	93.8	93.7	92.7	88.6	86.0
		MCWB	73.9	75.3	78.4	81.6	81.1	81.2	80.1	81.1	81.0	79.9	76.8	74.6
	5%	DB	84.6	88.5	92.7	96.4	100.8	99.3	93.8	92.1	92.1	91.1	87.4	84.8
		MCWB	73.5	75.4	78.3	81.2	82.1	81.2	80.5	80.9	81.1	79.6	76.5	73.6
	10%	DB	83.4	86.7	91.2	94.8	98.1	96.1	91.6	90.4	90.4	89.4	86.0	83.5
		MCWB	72.8	74.7	78.1	80.9	82.5	81.0	80.4	80.6	80.8	79.2	75.7	72.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.6	79.4	82.2	84.6	86.6	85.7	83.6	83.7	83.6	83.2	80.9	77.9
		MCDB	84.8	88.8	92.1	94.6	96.6	95.5	92.3	91.0	91.1	90.9	87.0	84.6
	2%	WB	76.0	78.0	80.9	83.6	85.3	84.5	82.6	82.6	82.7	81.9	79.6	76.6
		MCDB	83.1	87.0	90.7	93.3	94.9	94.0	90.5	89.8	90.1	89.3	85.5	83.2
	5%	WB	74.9	76.9	80.0	82.7	84.7	83.6	81.8	81.9	81.9	81.2	78.6	75.6
		MCDB	81.9	85.4	89.4	91.8	94.0	92.6	89.2	88.8	88.8	88.0	84.3	81.9
	10%	WB	73.9	75.9	79.2	81.9	84.0	82.7	81.1	81.2	81.4	80.4	77.7	74.6
		MCDB	81.0	83.8	88.0	90.7	93.1	91.2	88.2	87.8	87.8	86.8	83.3	80.9
Mean Daily Temperature Range	5% DB	MDBR	15.0	15.9	16.0	15.7	15.6	13.3	10.4	10.1	10.4	11.4	13.1	14.4
		MCDBR	16.0	18.3	17.5	17.0	19.4	18.8	14.2	12.9	13.5	14.3	14.5	15.3
		MCWBR	6.8	6.8	5.5	4.9	5.6	5.3	4.4	4.3	4.8	5.4	6.1	6.8
	5% WB	MCDBR	14.2	15.5	15.7	15.7	15.9	15.4	12.0	11.6	11.6	12.0	11.4	13.3
	MCWBR	6.8	6.6	5.8	5.7	6.4	6.1	4.9	4.8	5.0	5.4	5.5	6.8	
Clear-Sky Solar Irradiance	taub	0.514	0.506	0.575	0.584	0.643	0.633	0.578	0.558	0.555	0.552	0.526	0.540	
	taud	1.990	2.034	1.889	1.917	1.792	1.822	1.968	2.027	2.034	2.023	2.024	1.945	
	Ebn at Noon	244	254	241	239	222	222	235	242	243	239	239	231	
	Edh at Noon	54	54	65	63	70	68	59	56	55	54	52	56	
All-Sky Solar Radiation	RadAvg	1521	1783	1954	2065	1918	1478	1338	1362	1424	1463	1433	1383	
	RadStd	58	66	79	79	74	142	164	129	119	149	115	83	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) +0.95	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (3 neighbors)	+0.54	+0.54	N/A	N/A	+0.36	N/A	N/A	N/A	N/A	N/A